Preregistration for: Beliefs about sexual assault across time as a function of political ideology and identity

Summary

Provide a narrative summary of what is contained in this registration or how it differs from prior registrations. If this project contains documents for a preregistration, please note that here.

You may attach up to 5 file(s) to this question. You may attach files that you already have in OSF Storage in this project or upload (drag and drop) a new file from your computer. Uploaded files will automatically be added to this project so that they can be registered. To attach files from other components or an add-on, first add them to this project.

Background

Sexual assault remains a major problem that disproportionately affects women (RAINN, 2023). Social change movements, like #MeToo, have been paramount in raising awareness of the prevalence of sexual assault and in singling out perpetrators. However, the movement, as well as discussion about high-profile sexual assault allegations since, has spurred significant backlash from certain groups. After learning of a sexual assault allegation, these groups may, for instance, blame the victim as opposed to the perpetrator for the sexual assault that the victim experienced. These blame attributions have important downstream consequences for the well-being of victims and the success of movements like #MeToo in changing norms and policies around sexual assault.

The current research investigates the predictors of victim versus perpetrator blame across three critical moments in time revolving around sexual assault in the US. Specifically, we focus on the role of ideology, gender, historical context, threat, and other demographic characteristics on the tendency to blame the victim and blame the perpetrator.

This registration was written after the conclusion of data collection. The data were collected and preliminary analyses were conducted by the corresponding author. Both statistical consultants joined the project after all data collection was complete and intend to reclean and reanalyze the data. Thus this registration is being written before the reanalysis of the data has commenced. Although the corresponding author shared some of the preliminary results with all authors except the second statistical consultant, this registration is independent of those results. The second statistical consultant is blind to all prior analyses and findings. Below, we outline hypotheses, planned analyses, and exploratory analyses.

Method

Data were collected across three sessions:

- **First data collection: April 22 – May 21, 2017:** Before the #MeToo movement went viral in October 2017
- **Second data collection: February 26 – 28, 2018:** After the #MeToo movement went viral in October 2017 but before the Brett Kavanaugh Supreme Court hearings in September 2018
- **Third data collection: September 28 – 30, 2018:** After the Brett Kavanaugh Supreme Court hearings in September 2018 but before his confirmation in October 2018

Participants

Data from a total of 1,263 Amazon Mechanical Turk participants were collected across the three time points. The first group had 388 participants, the second group had 374 participants, and the third group had 501 participants. Respondents participated in only a single wave of data collection.

Measures

General facts vs. beliefs: To familiarize participants with the fact/belief questions (see below), participants were presented with 20 general statements and were asked to categorize the statements as facts or beliefs (e.g., "A hammer is used to pound nails"). These statements were also used as attention checks (Rabinowitz et al., 2013; see Data Cleaning section).

Threat condition: Participants were randomly assigned to write about either 3 or 10 real-life instances of threatening or fearful experiences (based on Thórisdóttir & Jost, 2011).

Feelings of threat: After writing about threatening or fearful experiences, participants were asked to rate on a scale from 1 to 10 how threatened they felt at the time (based on Thórisdóttir & Jost, 2011).

Victim blame: Participants were asked to report the degree to which they agreed or disagreed with 13 items blaming the victim for the sexual assault they experience (e.g., "If someone initiates kissing or hooking up, they can't be surprised if someone assumes they want to have sex;" 1 = Strongly disagree, 2 = Disagree, 3 = Agree, 4 = Strongly agree; based on Rabinowitz et al., 2013).

Predicted percentage of the adult population who blame the victim: Participants were asked to predict the percentage of the adult population who agree with each of the 13 items blaming the victim for the sexual assault they experience (0%, 25%, 50%, 75%, 100%; based on Rabinowitz et al., 2013).

Predicted percentage of men who blame the victim: Participants were asked to predict the percentage of men who agree with each of the 13 items blaming the victim for the sexual assault they experience (0%, 25%, 50%, 75%, 100%; based on Rabinowitz et al., 2013).

Predicted percentage of women who blame the victim: Participants were asked to predict the percentage of women who agree with each of the 13 items blaming the victim for the sexual assault they experience (0%, 25%, 50%, 75%, 100%; based on Rabinowitz et al., 2013).

Categorization of victim blame items as fact or belief: Participants were asked whether they thought each of the 13 items blaming the victim is a fact or a belief (based on Rabinowitz et al., 2013).

Perpetrator blame: Participants were asked the degree to which they agreed or disagreed with 13 items blaming the perpetrator for causing sexual assault (e.g., "It is sexual assault even if the act was consensual at the beginning, but the victim changed their mind at some point;" 1 = Strongly disagree, 2 = Disagree, 3 = Agree, 4 = Strongly agree; based on Rabinowitz et al., 2013).

Predicted percentage of the adult population who blame the perpetrator: Participants were asked to predict the percentage of the adult population who agree with each of the 13 items

blaming the perpetrator for causing sexual assault (0%, 25%, 50%, 75%, 100%; based on Rabinowitz et al., 2013).

Predicted percentage of men who blame the perpetrator: Participants were asked to predict the percentage of men who agree with each of the 13 items blaming the perpetrator for causing sexual assault (0%, 25%, 50%, 75%, 100%; based on Rabinowitz et al., 2013).

Predicted percentage of women who blame the perpetrator: Participants were asked to predict the percentage of women who agree with each of the 13 items blaming the perpetrator for causing sexual assault (0%, 25%, 50%, 75%, 100%; based on Rabinowitz et al., 2013).

Categorization of perpetrator blame items as fact or belief: Participants were asked whether they thought each of the 13 items blaming the perpetrator is a fact or a belief (based on Rabinowitz et al., 2013).

General political ideology: Participants were asked their political ideology (Liberal, Moderate, Conservative, Other (Please specify)).

Economic ideology: Participants were asked how they would rate their views on economic issues (1 = Strongly liberal, 7 = Strongly conservative, or Other (Please specify)).

Social ideology: Participants were asked how they would rate their views on social issues (1 = Strongly liberal, 7 = Strongly conservative, or Other (Please specify)).

Race: Participants were asked to indicate their racial identification (American Indian or Alaskan Native, Asian, Black or African American, Hispanic or Latino, Native Hawaiian or other Pacific Islander, White (including Middle Eastern), Other (please specify)).

First Language: Participants were asked whether English is their first language (Yes, No).

Participant Sex: Participants were asked to indicate their sex (Female, Male, Other).

Age: Free response

Sexual orientation: Participants were asked to indicate their sexual orientation (Homosexual/Lesbian, Heterosexual, Bisexual, Asexual, Other (Please specify)).

Religion: Participants were asked to indicate their religion (free response).

Religiosity: Participants were asked how strongly they identify with and practice their religion (0 = Not at all, 4 = Fully)

A. Preliminary test and hypothesis

1. <u>Manipulation check</u>

Model:

We will estimate an independent samples *t*-test in which threat condition (3, 10) will serve as the predictor and feelings of threat will serve as the outcome.

Research question:

Did writing about 3 (vs. 10) threatening experiences lead to greater feelings of threat?

If the manipulation check is significant, threat condition will be included as a moderator in all models in which feelings of threat are estimated.

B. Confirmatory tests and hypotheses

1. Effect of threat and target on blame within and across session among liberal and conservative males and females

Model:

We will estimate a 2 (Target: victim, perpetrator) X 2 (Participant sex: male, female) X 3 (General political ideology: liberal, moderate, conservative) X 3 (Session: first, second, third) X continuous (Feelings of threat) multilevel model. Target will serve as a within-subject factor; participant sex, general political ideology, session, and feelings of threat will serve as between-subjects factors; and blame score (see Data Cleaning section) will serve as the outcome. Age, sexual orientation, and religiosity will be entered as covariates. Reduced models will be examined as robustness checks. Models using social rather than general political ideology will be examined as robustness checks.

Hypotheses:

Participants will blame the perpetrator more than they blame the victim (main effect of target). Males will blame the victim more than will females; Females will blame the perpetrator more than will males (target x participant sex interaction). Liberals will blame the perpetrator more than moderates and conservatives; Conservatives will blame the victim more than moderates and liberals (target x general political ideology interaction). The tendency to blame the victim will increase and the tendency to blame the perpetrator will decrease from session 1 to session 2, and from session 2 to session 3 (we also expect this pattern when comparing session 1 to session 3; target x session interaction); especially among men (target x participant sex x session interaction), conservatives (target x general political ideology x session interaction), and conservative men (target x participant sex x general political ideology x session interaction). Feelings of threat will be positively associated with victim blame and negatively associated with perpetrator blame (target x feelings of threat interaction). All other comparisons are exploratory. We expect that these effects will hold adjusting for age, sexual orientation, and religiosity.

2. Threat within and across session among liberal and conservative males and females

Model:

We will estimate a 2 (Participant sex: male, female) X 3 (General political ideology: liberal, moderate, conservative) X 3 (Session: first, second, third) between-subjects ANCOVA. Feelings of threat will serve as the outcome variable. Age, sexual orientation, and religiosity will be entered as covariates. Reduced models will be examined as robustness checks.

Regression models using social rather than general political ideology will be examined as robustness checks.

Hypotheses:

Males will report greater feelings of threat than will females (main effect of participant sex). Conservatives will report greater feelings of threat than will moderates and liberals (main effect of general political ideology). Conservative males will report the highest feelings of threat (participant sex x general political ideology interaction). Feelings of threat will increase from session 1 to session 2, and from session 2 to session 3 (and from session 1 to session 3; main effect of session); especially among men (participant sex x session interaction), conservatives (general political ideology x session interaction), and conservative men (participant sex x general political ideology x session interaction). All other comparisons are exploratory. We expect that these effects will hold adjusting for age, sexual orientation, and religiosity.

3. **Mediation of blame by threat across session among liberal and conservative males and females**

Model:

We will estimate a moderated mediation model. Session (first, second, third) will serve as the predictor, feelings of threat (continuous) will serve as the mediator, and blame will serve as the outcome. General political ideology (liberal, moderate, conservative) and participant sex (male, female) will serve as the moderators on the a and c paths and target will serve as the moderator on the b and c paths. Age, sexual orientation, and religiosity will be entered as covariates on a and c paths. Reduced models will be examined as robustness checks. Models using social rather than general political ideology will be examined as robustness checks.

Hypotheses:

The association between session and blame (c path) will be moderated by target, such that session will be positively associated with blame when the target is the victim, and negatively associated with blame when the target is the perpetrator (target x session interaction on blame). These effects will be mediated by greater feelings of threat. Participant sex and general political ideology will moderate the a and c paths such that the association between session and feelings of threat and session and blame will be strongest for males (participant sex x session on threat; target x participant sex x session on blame), conservatives (general political ideology x session on threat; target x general political ideology x session on blame), and male conservatives (participant sex x general political ideology x session on threat; target x participant sex x general political ideology x session on blame). The association between feelings of threat and blame (b path) will be moderated by target, such that threat will be positively associated with blame when the target is the victim, and negatively associated with blame when the target is the perpetrator (target x feelings of threat interaction on blame). All other comparisons are exploratory. We expect that these effects will hold adjusting for age, sexual orientation, and religiosity.

C. Exploratory tests and research questions

1. **Categorization of blame items as facts vs. beliefs across session among liberal and conservative males and females**

Model:

We will estimate a 2 (Target: victim, perpetrator) X 2 (Participant sex: male, female) X 3 (General political ideology: liberal, moderate, conservative) X 3 (Session: first, second, third) mixed-model ANCOVA. Target will serve as a within-subject factor; participant sex, general political ideology, and session will serve as between-subjects factors; and participants' average categorization of the items as fact (vs. beliefs; see Data Cleaning section) will serve as the outcome variable. Age, sexual orientation, and religiosity will be entered as covariates. Reduced models will be examined as robustness checks. Models using social rather than general political ideology will be examined as robustness checks.

Research question:

What is the effect of target, participant sex, general political ideology, session, and their interactions on the categorization of the blame items as fact or belief?

2. **Predictions about other men's and women's agreement with them on victim and perpetrator blame across session among liberal and conservative males and females**

Model:

We will estimate a 2 (Target: victim, perpetrator) X 3 (Predicted group: population, men, women) X 2 (Participant sex: male, female) X 3 (General political ideology: liberal, moderate, conservative) X 3 (Session: first, second, third) mixed-model ANCOVA. Target and predicted group will serve as within-subject factors; participant sex, general political ideology, and session will serve as between-subjects factors; and predicted percentage will serve as the outcome. Age, sexual orientation, and religiosity will be entered as covariates. Reduced models will be examined as robustness checks.

Research question:

What is the effect of target, predicted group, participant sex, general political ideology, session, and their interactions on participants' predictions regarding the percent of other individuals, men, and women who endorse the blame items?

3. **Assumed similarity with other men and women on victim and perpetrator blame across session among liberal and conservative males and females**

Model:

We will estimate a 2 (Target: victim, perpetrator) X 3 (Predicted group: population, men, women) X 2 (Participant sex: male, female) X 3 (General political ideology: liberal, moderate, conservative) X 3 (Session: first, second, third) mixed-model ANCOVA. Target and predicted group will serve as within-subject factors; participant sex, general political ideology, and session will serve as between-subjects factors; and assumed similarity will serve as the outcome variable (see Data Cleaning section). Age, sexual orientation, and

religiosity will be entered as covariates. Reduced models will be examined as robustness checks.

Research question:

What is the effect of target, predicted group, participant sex, general political ideology, session, and their interactions on the degree to which one's own responses on the blame items are associated with predictions about the population, women's, and men's agreement with them?

4. **Accuracy in assumed similarity across session among liberal and conservative males and females**

Model:

We will estimate a 2 (Target: victim, perpetrator) X 3 (Predicted group: population, men, women) X 2 (Participant sex: male, female) X 3 (Ideology, liberal, moderate, conservative) X 3 (Session: first, second, third) mixed-model ANCOVA. Target and the predicted group will serve as the within-subject factors; participant sex, general political ideology, and session will serve as the between-subjects factors; and accuracy will serve as the outcome (see Data Cleaning section). Age, sexual orientation, and religiosity will be entered as covariates. Reduced models will be examined as robustness checks. We will also estimate one-sample *t*-tests within each subgroup (males, females, conservatives, moderates, liberals) to examine whether accuracy for population, men, and women differs significantly from 0.

Research question:

How does target, predicted group, participant sex, general political ideology, session, and/or their interactions influence the degree to which participants' own responses to the blame items accurately inform their inferred agreement with others in the population, men, and women?

5. **Actual fact/belief consensus with other men and women on victim and perpetrator blame across session among liberal and conservative males and females**

Model

We will estimate two Social Relations Models using logistic generalized linear mixed models (one for victim blame and one for perpetrator blame) to partition variance due to target (i.e., agreement among participants about whether items are facts or beliefs) vs. perceiver (i.e., consistency within participants in rating items as facts or beliefs) on categorization of blame items as fact vs. belief. We will estimate consensus as a function of absolute target variance (i.e., whether variance attributable to item agreement is significantly different from 0). We will estimate the magnitude of the consensus by calculating the proportional change in the pseudo-R^2 due to target in this model compared to a model with no random effects. Session, age, sexual orientation, and religiosity will be entered as covariates. Reduced models will be examined as robustness checks.

We will also conduct subgroup analyses for males, females, liberals, moderates, and conservatives.

Research question:

What degree of actual consensus exists within male, female, liberal, moderate, and conservative subgroups on categorization of the victim and perpetrator blame items as fact or belief?

6. **Actual blame consensus with other men and women on victim and perpetrator blame across session among liberal and conservative males and females**

Model:

We will estimate two Social Relations Models (one for victim blame and one for perpetrator blame) to partition variance due to the target (i.e., agreement among participants on blame items), perceiver (i.e., consistency within participants on items), and error (target x perceiver interactions and additional sources of unexplained error, or differences in patterns of blame across participants) on blame. We will estimate consensus as a function of absolute target variance (i.e., whether variance attributable to item agreement is significantly different from 0). We will estimate the magnitude of the consensus by calculating the proportion of total variance explained by target. Session, age, sexual orientation, and religiosity will be entered as covariates. Reduced models will be examined as robustness checks.

We will also conduct subgroup analyses for males, females, liberals, moderates, and conservatives.

Research question:

What degree of actual consensus exists within male, female, liberal, moderate, and conservative subgroups on victim and perpetrator blame?

D. Data Cleaning

Exclusion criteria:

To filter out poor responses, we will implement a "flagging" system in which participants will receive flags for poor responses:

Participants who report that English is not their first language will receive a flag. Participants who took more than an hour or less than 10 minutes to complete the survey will receive a flag (the survey was expected to have taken participants ~30 minutes to complete). Participants who have no variability in answers within the scales we believe should generally have variability (i.e., those with reverse-coded items) will receive a flag. Participants with no variability in one of these scales will receive one flag. Participants with low variability on two or more of these scales will receive two flags. Participants who said that a fact is a belief or that a belief is a fact in the general fact/belief task will receive a flag (for instance, if they indicated that "A hammer is used to pound nails" is a belief; Rabinowitz et al., 2013). Participants who wrote open-ended responses using poor English or who wrote about

something unrelated to the prompt will receive between 1 and 4 flags depending on the quality of the answer.

Prior to conducting any analyses, we will determine a flag cutoff point. Participants whose flags and/or free responses indicate clear failure to follow study instructions or insincere responses will not be further analyzed.

Robustness checks:
Upon final analysis, we will conduct robustness checks. If the patterns of data found while excluding poor respondents do not change when including them, we will report the results using the larger sample. If they do change any results, we will exclude them from the report and indicate the exclusion rationale in any resulting manuscript.

Data transformation:

Victim blame: We will calculate the reliability of the 13 victim blame items. If the Cronbach's α is .70 or greater, we will average all items to create an aggregate index of victim blame. If the α is lower than .70, and/or we identify problematic items (e.g., items that are negatively correlated with several other items; items that would substantially improve α if they were removed), we will determine how to utilize this scale prior to testing any research hypotheses. If possible, we will remove problematic items and recalculate an aggregate score. If, however, reliability is quite poor, we will treat victim blame as a repeated measure in all analyses rather than creating an aggregate index.

Perpetrator blame: We will calculate the reliability of the 13 perpetrator blame items. If the Cronbach's α is .70 or greater, we will average all items to create an aggregate index of perpetrator blame. If the α is lower than .70, and/or we identify problematic items (e.g., items that are negatively correlated with several other items; items that would substantially improve α if they were removed), we will determine how to utilize this scale prior to testing any research hypotheses. If possible, we will remove problematic items and recalculate an aggregate score. If, however, reliability is quite poor, we will treat perpetrator blame as a repeated measure in all analyses rather than creating an aggregate index.

Categorization of victim blame items as fact or belief: We will create an average index of participants' categorization of the 13 victim blame items as fact or belief.

Categorization of perpetrator blame items as fact or belief: We will create an average index of participants' categorization of the 13 perpetrator blame items as fact or belief.

Assumed similarity (with the population, women, and men on victim blame and perpetrator blame): We will calculate scores for each participant that represents the degree to which the participant believes that others' attitudes are similar to their own. To do so, we will correlate participants' response on each of the blame items with their predicted agreement score on each item (i.e., participants' predictions regarding the percent of others who agree with each item). We will create six variables. The first three will correlate participants' responses on each of the 13 victim blame items with their predicted agreement among the population (using their predicted percentage of the population who blame the victim on each of the 13 victim blame items), women (predicted percentage of women who blame the victim on each of the 13 victim blame items), and men (predicted percentage of men who blame the victim on each of the 13 victim blame items). The second three assumed similarity scores will

correlate participants' responses on each of the 13 perpetrator blame items with their predicted agreement among the population (using their predicted percentage of the population who blame the perpetrator on each of the 13 perpetrator blame items), men (predicted percentage of men who blame the perpetrator on each of the 13 perpetrator blame items), and women (predicted percentage of women who blame the perpetrator on each of the 13 perpetrator blame items). We will then *r*-to-*z* transform each correlation to unbound it.

Robustness checks:
We will conduct a series of additional robustness checks for models examining assumed similarity. First, because participants reported their own opinions on a 4-point scale (Strongly disagree, disagree, agree, strongly agree) but were asked to estimate the percentage of others who "agree" (vs. disagree), we will reanalyze our model following the strategy used by Kruger & Clement (1994). In particular, responses on the blame items of "Strongly disagree" and "disagree" will be combined as "disagree" and responses of "agree" and "strongly agree" will be combined as "agree".

Second, because this variable requires within-person variability on both participants' own responses across items and their predicted agreement estimates, participants with no variability are dropped from analysis. This may be overly restrictive, so we will reanalyze our model after slightly jittering the responses of participants with 0 variability on one or both variables.

Finally, because the measure of assumed similarity is extremely sensitive to mild variations in responses, we will estimate a third alternate model. Specifically, we will estimate a 2 (Target: victim, perpetrator) X 3 (Predicted group: population, men, women) X 2 (Participant sex: male, female) X 3 (General political ideology: liberal, moderate, conservative) X 3 (Session: first, second, third) X continuous (aggregate victim blame) X continuous (aggregate perpetrator blame) multilevel model. Target and predicted group will serve as within-subject factors; participant sex, general political ideology, session, aggregate victim blame, and aggregate perpetrator blame (see Data Cleaning section) will serve as between-subject factors; and predicted percentage will serve as the outcome variable. Age, sexual orientation, and religiosity will be entered as covariates. Reduced models will be examined as robustness checks.

We will report the findings of these alternate analyses in a footnote or Supplement.

Accuracy (with the population, men, and women about similarity in victim blame and perpetrator blame): We will calculate scores for each participant that represents their accuracy in assumed similarity. To do so, we will a) calculate the percentage of participants who actually agreed with each item by dichotomizing each participant's blame scores such that "strongly disagree" and "disagree" are coded as "disagree," and "strongly agree" and "agree" are coded as "agree". We will then b) calculate a discrepancy score for each participant on each item by subtracting the percentage of people (population, men, and women) who agreed with the item (calculated in step a) from the participant's estimate of the percentage of people who agreed with each item. We will then c) correlate the participant's victim and perpetrator blame original, unaltered scores with the discrepancy score. Lastly, d) we will *r*-to-*z* transform each correlation to unbound it.

Robustness checks:

We will conduct additional robustness checks for models examining accuracy. Again following Kruger & Clement (1994), we will reanalyze our model using the binary version of the blame scores in step c rather than the original, unaltered scores.

Second, because this variable again requires within-person variability, we will reanalyze our model after slightly jittering responses of participants with 0 variability.

We will report the findings of these alternate analyses in a footnote or Supplement.

Demographics recodes:

Ideology: Participants who identify as "Other" for general, economic, or social ideology will be recoded as missing on these variables.

Sex: Participants who do not identify as male or female (i.e. "other") will be recoded as missing on this variable.

Sexual orientation: Categories with fewer than ten participants will be grouped into an "other" category. If there is only one category with fewer than ten participants, those participants will be recoded as missing on this variable.

References:

Rabinowitz, M., Acevedo, M., Casen, S., Rosengarten, M., Kowalczyk, M., & Portnoy, L. B. (2013). Distinguishing facts from beliefs: Fuzzy categories. *Psychology of Language and Communication*, *17*(3), 241-268.

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Thórisdóttir, H., & Jost, J. T. (2011). Motivated closed-mindedness mediates the effect of threat on political conservatism. *Political Psychology*, *32*(5), 785-811.